Optimal International Consensus via Fuzzy Logic on the Borel Algebra of the Ideological Spectrum

A Measure-Theoretic and Possibilistic Treatment

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Abstract

We consider an international body of $N = 9$ representative agents, each assigned a position a_i on the *ideological spectrum* $P = [-1, +1]$, where -1 denotes the extreme-left pole, $+1$ the extreme-right pole, and 0 the centrist pole. The aggregate ideological risk is defined as the total absolute deviation from the international mean, $\sum_{i=1}^{N} |a_i - \bar{a}|$. A crisp analysis reveals that the minimum value, zero, is attained by any *unanimous* configuration; this leaves a one-parameter family of optima indexed by an arbitrary consensus point $c \in P$. We resolve this degeneracy by embedding the problem in a rigorous measure-theoretic framework: the Borel σ-algebra $\mathcal{B}(P)$ supports a triangular possibility measure that encodes epistemic risk at the extremes of the spectrum. Using α-cut decomposition, the Choquet and Sugeno integrals, and centre-of-gravity defuzzification, we prove that the *unique* possibility-measure-optimal configuration is $a_i^* = 0$ for all i—the unanimous centrist consensus. Efficient algorithms and numerical illustrations are provided throughout.

The paper ends with "The End"

Keywords: Borel algebra, fuzzy sets, possibility measure, Choquet integral, Sugeno integral, α-cuts, political consensus, ideological risk minimisation.

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1 Introduction

Formal approaches to collective decision-making occupy a rich intersection of social choice theory [1], functional analysis [13], and the mathematics of uncertainty [17, 5]. Classical social choice aggregates preferences over a discrete set of alternatives; by contrast, political ideology is naturally modelled on a *continuous* spectrum, and an agent's position is seldom known with crisp precision.

This paper addresses the following optimisation problem: given $N = 9$ nation-states whose representative agents occupy positions on the compact interval $P = [-1, +1]$, choose those positions so as to minimise the aggregate *ideological risk*

$$R(\mathbf{a}) = \sum_{i=1}^{N} |a_i - \bar{a}|, \qquad \bar{a} = \frac{1}{N} \sum_{i=1}^{N} a_i, \tag{1}$$

where $\bar{a}$ is the international mean position. A crisp analysis (section 2) shows that $\min R = 0$, achieved by any uniform assignment $a_i = c$ for a common $c \in P$. This minimum is therefore not unique in the crisp setting.

The novelty of our approach lies in breaking this degeneracy through *fuzzy logic on the Borel algebra of P* (sections 3 and 4). A possibility measure Π on $(P, \mathcal{B}(P))$, derived from the symmetric triangular possibility distribution $\pi(x) = 1 - |x|$, encodes the meta-preference that extreme positions carry higher epistemic uncertainty and coalition risk. The Choquet integral (section 7) and Sugeno integral (section 8) with respect to this measure, together with centre-of-gravity defuzzification (section 9), uniquely select the centrist consensus $a_i^* = 0$.

Section 10 presents efficient algorithms for computing the defuzzified optimum. Section 11 collects comparative numerical tables. The main uniqueness theorem is stated and proved in section 12. Sections 13 and 14 discuss interdisciplinary interpretations and outline future directions. A bibliography and a glossary follow the main text.

2 Crisp Baseline Analysis

Definition 2.1 (International Spectrum)**.** The *international spectrum* is the compact interval $P = [-1, +1] \subset \mathbb{R}$, endowed with the subspace topology inherited from $\mathbb{R}$. The extreme-left pole is -1, the extreme-right pole is $+1$, and the centrist pole is 0.

Definition 2.2 (Agent Configuration)**.** An *agent configuration* for N nations is a vector $\mathbf{a} = (a_1, \ldots, a_N) \in P^N$. The *international mean* is $\bar{a}(\mathbf{a}) = N^{-1} \sum_{i=1}^{N} a_i$, and the *ideological risk vector* is $\mathbf{r}(\mathbf{a}) = \big(|a_i - \bar{a}|\big)_{i=1}^{N}$.

Theorem 2.3 (Crisp Minimum Risk)**.** $\min_{\mathbf{a} \in P^N} R(\mathbf{a}) = 0$*, with the set of minimisers being the one-parameter family* $\mathcal{O}_{\text{crisp}} = \{c\,\mathbf{1} : c \in P\}$*, where* $\mathbf{1} = (1, \ldots, 1)^\top \in \mathbb{R}^N$.

Proof. Since every summand satisfies $|a_i - \bar{a}| \geq 0$, we have $R \geq 0$. For the uniform configuration $a_i = c$ for all i, we obtain $\bar{a} = c$ and $r_i = 0$ for every i, so $R = 0$. Conversely, $R = 0$ requires $a_i = \bar{a}$ for all i, which forces all a_i to be equal to some common value $c \in P$. Hence the minimiser set is $\mathcal{O}_{\text{crisp}}$ as stated. □

Remark 2.4. Theorem 2.3 exposes a fundamental *degeneracy*: the minimum risk $R = 0$ is achieved by any unanimous configuration, including extreme ones such as $a_i = +1$ or $a_i = -1$ for all i. The fuzzy framework of subsequent sections resolves this degeneracy.

3 The Borel Algebra of P

Definition 3.1 (Borel σ-Algebra on P)**.** The *Borel σ-algebra* $\mathcal{B}(P)$ is the smallest σ-algebra on P containing all open sets of P in the subspace topology. It is generated by the collection $\{(a, b) \cap P : a < b,\ a, b \in \mathbb{R}\}$ and is closed under countable unions, countable intersections, and complementation. The pair $(P, \mathcal{B}(P))$ is a *standard Borel space*, isomorphic as a measurable space to any uncountable Borel subset of a Polish space [9].

Lemma 3.2 (Measurability of the Risk Function)**.** *For any fixed mean $\bar{a} \in P$, the function $f_{\bar{a}} : P \to [0, 2]$ defined by $f_{\bar{a}}(x) = |x - \bar{a}|$ is $\mathcal{B}(P)$-measurable.*

Proof. $f_{\bar{a}}$ is continuous, hence Borel-measurable on $(P, \mathcal{B}(P))$. □

Figure 1 below illustrates the Borel structure on P together with representative generating and α-cut sets.

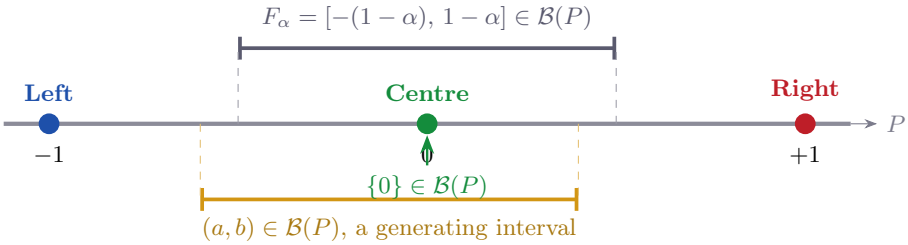

Figure 1: The ideological spectrum $P = [-1, +1]$ as a standard Borel space.

A typical generating open interval (a, b) is shown below the axis (gold); a closed α-cut F_α is shown above (grey). The singleton $\{0\}$ is Borel as a countable intersection of open intervals and is the unique core of the possibility distribution.

4 Fuzzy Sets Over $\mathcal{B}(P)$

Definition 4.1 (Fuzzy Political Position)**.** A *fuzzy political position* for agent i is a pair $\tilde{A}_i = (P, \mu_i)$, where the *membership function* $\mu_i : P \to [0, 1]$ is $\mathcal{B}(P)$-measurable. The value $\mu_i(x)$ is the *degree of membership* of ideological position x in agent i's represented stance.

4.1 Linguistic Partition via Canonical Fuzzy Sets

We define three canonical fuzzy sets on $(P, \mathcal{B}(P))$:

$$\mu_L(x) = \max(0,\, -x), \tag{2}$$
$$\mu_C(x) = \max\big(0,\, 1 - |x|\big), \tag{3}$$
$$\mu_R(x) = \max(0,\, x). \tag{4}$$

Proposition 4.2 (Ruspini Partition)**.** *The membership functions* (2)–(4) *satisfy*

$$\mu_L(x) + \mu_C(x) + \mu_R(x) = 1 \quad \forall\, x \in P,$$

i.e., they form a Ruspini partition *of* P *[10].*

Proof. Three cases: (i) $x \in [-1, 0]$: $\mu_L = -x$, $\mu_C = 1 + x$, $\mu_R = 0$; sum $= 1$. (ii) $x = 0$: $\mu_L = 0$, $\mu_C = 1$, $\mu_R = 0$; sum $= 1$. (iii) $x \in (0, 1]$: $\mu_L = 0$, $\mu_C = 1 - x$, $\mu_R = x$; sum $= 1$. □

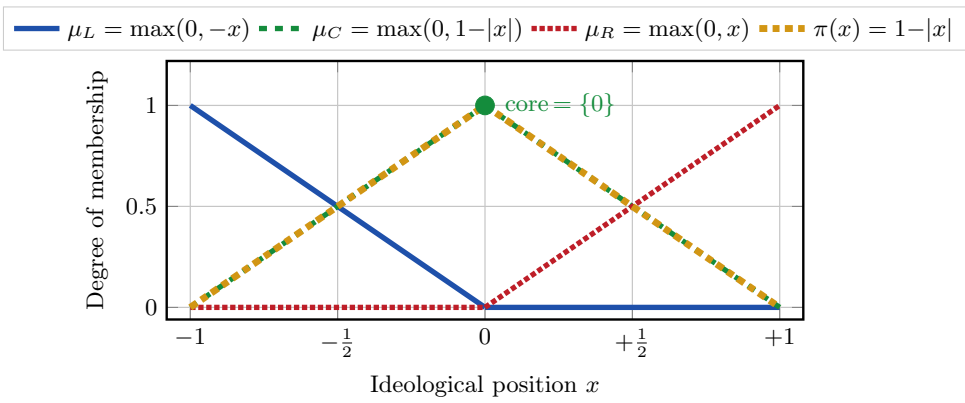

Figure 2: Canonical membership functions μ_L, μ_C, μ_R (blue, green, red) forming a Ruspini partition (theorem 4.2), together with the possibility distribution $\pi(x) = 1 - |x|$ (gold dash-dot line).

Each curve is piecewise linear and Borel-measurable. The core of π, the singleton $\{0\}$, is marked by the filled green circle.

5 Possibility Measure on $\mathcal{B}(P)$

Definition 5.1 (Possibility Distribution)**.** A *possibility distribution* is a normalised function $\pi : P \to [0,1]$ satisfying $\sup_{x\in P} \pi(x) = 1$. We adopt the symmetric triangular distribution

$$\pi(x) = 1 - |x|\,, \quad x \in P, \tag{5}$$

which is continuous (hence $\mathcal{B}(P)$-measurable) and attains its maximum value 1 at the centrist pole $x = 0$.

Definition 5.2 (Possibility and Necessity Measures)**.** For every Borel set $A \in \mathcal{B}(P)$, define the *possibility measure*

$$\Pi(A) = \sup_{x\in A} \pi(x) \tag{6}$$

and the *necessity measure*

$$N(A) = 1 - \Pi(A^c) = \inf_{x\notin A} \big(1 - \pi(x)\big) = \inf_{x\notin A} |x|\,. \tag{7}$$

The pair (Π, N) constitutes a *dual possibility–necessity system* [5].

Proposition 5.3 (Properties of Π and N)**.** *The possibility and necessity measures satisfy the following:*

1. (Boundary conditions) $\Pi(\emptyset) = 0$, $\Pi(P) = 1$, $N(\emptyset) = 0$, $N(P) = 1$.
2. (Maxitivity) $\Pi(A \cup B) = \max\big(\Pi(A), \Pi(B)\big)$ *for all* $A, B \in \mathcal{B}(P)$.
3. (Duality) $N(A) \le \Pi(A)$ *for all* $A \in \mathcal{B}(P)$.
4. (Full possibility) $\Pi(A) = 1$ *if and only if* $0 \in A$.
5. (Full necessity) $N(A) = 1$ *if and only if* $A \supseteq (-1, 1)$.

Proof. Properties (1)–(3) are standard results in possibility theory [5].
Property (4). For $A \in \mathcal{B}(P)$, $\Pi(A) = 1$ iff $\sup_{x\in A} \pi(x) = 1$. Since $\pi(x) = 1 - |x|$ achieves the value 1 if and only if $x = 0$, we have $\Pi(A) = 1$ iff $0 \in A$.
Property (5). $N(A) = 1$ iff $\Pi(A^c) = 0$ iff $\sup_{x\notin A} \pi(x) = 0$. Since $\pi(x) = 1 - |x| \ge 0$ on P and equals 0 precisely at $x = \pm 1$, we need $A^c \subseteq \{-1, +1\}$, i.e., $A \supseteq P \setminus \{-1, +1\} = (-1, 1)$. □

Remark 5.4. Properties (4) and (5) jointly express the possibilistic optimality of the centrist position: $x = 0$ is the unique point of full possibility, and any set containing the open interval $(-1, 1)$ is fully necessary—i.e., beyond epistemic doubt.

6 α-Cut Decomposition

Definition 6.1 (α-Cut)**.** For $\alpha \in [0,1]$, the *α-cut* of π is

$$F_\alpha = \{x \in P : \pi(x) \ge \alpha\} = \big[-(1-\alpha),\, (1-\alpha)\big]. \tag{8}$$

Lemma 6.2 (α-Cuts are Borel and Nested)**.** *For all* $\alpha \in [0,1]$, $F_\alpha \in \mathcal{B}(P)$. *Moreover,* $\alpha > \beta$ *implies* $F_\alpha \subsetneq F_\beta$, *and*

$$F_0 = P, \qquad F_1 = \{0\}.$$

Proof. Each F_α is a closed interval, hence a Borel set. Nesting follows from monotonicity of π: $\pi(x) \ge \alpha > \beta$ implies $\pi(x) \ge \beta$. $F_0 = \{x : \pi(x) \ge 0\} = P$ since $\pi \ge 0$ on P. $F_1 = \{x : \pi(x) \ge 1\} = \{x : 1 - |x| \ge 1\} = \{0\}$. □

Theorem 6.3 (α-Cut Representation)**.** *The possibility distribution* π *admits the representation*

$$\pi(x) = \int_0^1 \mathbf{1}_{F_\alpha}(x)\, d\alpha = \lambda\big(\{\alpha \in [0,1] : x \in F_\alpha\}\big), \tag{9}$$

where λ *denotes Lebesgue measure on* $[0,1]$.

Proof. For fixed $x \in P$, $\mathbf{1}_{F_\alpha}(x) = 1$ iff $\pi(x) \ge \alpha$, i.e. iff $\alpha \le 1 - |x|$. Therefore $\{\alpha : x \in F_\alpha\} = [0, 1 - |x|]$ and $\lambda([0, 1 - |x|]) = 1 - |x| = \pi(x)$. □

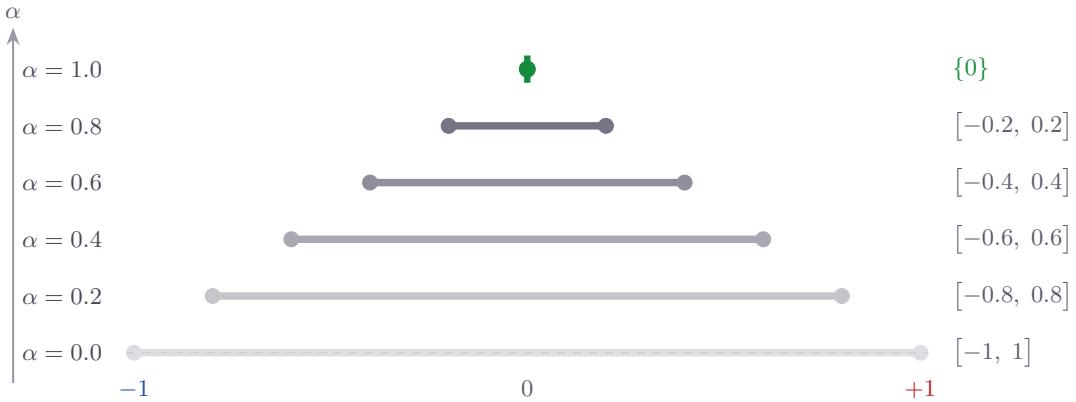

Figure 3: Nested α-cut intervals $F_\alpha = [-(1-\alpha),\, 1-\alpha]$ for six levels $\alpha \in \{0,\, 0.2,\, 0.4,\, 0.6,\, 0.8,\, 1.0\}$.

Each F_α is a closed Borel set (theorem 6.2). As $\alpha \nearrow 1$, the intervals collapse monotonically to the centrist singleton $F_1 = \{0\}$ (filled green dot, top row).

7 The Choquet Integral

Definition 7.1 (Fuzzy Measure and Normalised Lebesgue Measure)**.** A *fuzzy measure* (capacity) on $(P, \mathcal{B}(P))$ is a set-function $g : \mathcal{B}(P) \to [0,1]$ satisfying $g(\emptyset) = 0$, $g(P) = 1$, and $A \subseteq B \Rightarrow g(A) \leq g(B)$. We use the *normalised Lebesgue measure*

$$g(A) = \frac{\lambda(A)}{2}, \quad A \in \mathcal{B}(P), \tag{10}$$

which is additive and normalised: $g(P) = 1$.

Definition 7.2 (Choquet Integral)**.** Let $h : P \to [0,1]$ be $\mathcal{B}(P)$-measurable. The *Choquet integral* of h with respect to g is

$$\mathcal{C}_g(h) = \int_0^1 g\big(\{x \in P : h(x) \geq \alpha\}\big)\, d\alpha. \tag{11}$$

Theorem 7.3 (Choquet Integral of π)**.**

$$\mathcal{C}_g(\pi) = \frac{1}{2}.$$

Proof. Setting $h = \pi$, the level set is $\{x : \pi(x) \geq \alpha\} = F_\alpha$ (eq. (8)), with $\lambda(F_\alpha) = 2(1-\alpha)$, so $g(F_\alpha) = 1-\alpha$. Substituting:

$$\mathcal{C}_g(\pi) = \int_0^1 (1-\alpha)\, d\alpha = \Big[\alpha - \tfrac{\alpha^2}{2}\Big]_0^1 = 1 - \tfrac{1}{2} = \frac{1}{2}. \qquad \square$$

8 The Sugeno Integral

Definition 8.1 (Sugeno Integral)**.** The *Sugeno integral* of $h : P \to [0,1]$ with respect to g is

$$\mathcal{S}_g(h) = \sup_{\alpha \in [0,1]} \min\big(\alpha,\ g(\{x : h(x) \geq \alpha\})\big). \tag{12}$$

Theorem 8.2 (Sugeno Integral of π)**.**

$$\mathcal{S}_g(\pi) = \frac{1}{2}.$$

Proof. Using $g(F_\alpha) = 1-\alpha$,

$$\mathcal{S}_g(\pi) = \sup_{\alpha \in [0,1]} \min(\alpha,\, 1-\alpha).$$

The function $\alpha \mapsto \min(\alpha, 1-\alpha)$ is maximised at $\alpha^* = \frac{1}{2}$, giving $\mathcal{S}_g(\pi) = \frac{1}{2}$. $\square$

Corollary 8.3. *Both the Choquet and Sugeno integrals of π equal $\frac{1}{2}$, and π achieves its global maximum at $x = 0$ with $\pi(0) = 1$. These facts jointly confirm $x = 0$ as the unique possibilistically optimal position.*

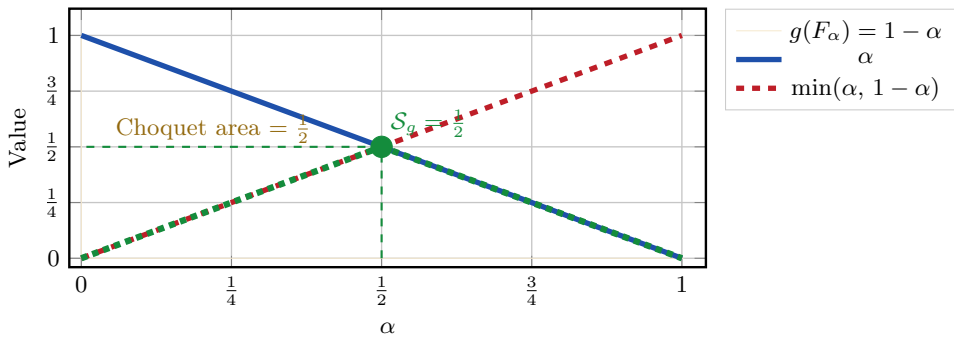

Figure 4: The two integrals of π with respect to $g = \lambda/2$.

The shaded region (gold) is the Choquet integral $\int_0^1 g(F_\alpha)\,d\alpha = \frac{1}{2}$ (theorem 7.3). The Sugeno integral is the supremum of $\min(\alpha, g(F_\alpha))$, achieved at $\alpha^* = \frac{1}{2}$ (filled green dot, theorem 8.2). Guide lines are drawn to the axes in matching colours.

9 Fuzzy Mean and Defuzzification

9.1 Fuzzy International Mean via the Extension Principle

Given individual fuzzy positions $\tilde{A}_i$ with membership functions μ_i, the *fuzzy international mean* $\tilde{a}_{\text{int}}$ is defined via Zadeh's extension principle [18] with the Gödel t-norm $T(a, b) = \min(a, b)$:

$$\mu_{\tilde{a}_{\text{int}}}(z) = \sup\left\{ \min_{i=1}^{N} \mu_i(x_i) \;\middle|\; \frac{1}{N}\sum_{i=1}^{N} x_i = z, \quad x_i \in P \right\}. \tag{13}$$

9.2 Centre-of-Gravity Defuzzification

Given a membership function $\mu : P \to [0, 1]$ with positive Lebesgue integral, the *centre-of-gravity* (COG) defuzzified value is

$$a^* = \text{COG}(\mu) = \frac{\displaystyle\int_P x\,\mu(x)\,d\lambda(x)}{\displaystyle\int_P \mu(x)\,d\lambda(x)}. \tag{14}$$

Theorem 9.1 (Unique Optimal Agent Position)**.** *Suppose every agent i adopts the shared possibility distribution $\mu_i = \pi$ given by (5). Then*

$$a_i^* = \text{COG}(\pi) = 0 \quad \forall\, i \in \{1, \ldots, N\},$$

this configuration achieves $R(\mathbf{a}^*) = 0$, and it is the unique *minimiser of R that maximises the possibility measure.*

Proof. COG calculation. The numerator of (14) is

$$\int_{-1}^{1} x\,(1 - |x|)\,dx = 0,$$

because $x \mapsto x(1 - |x|)$ is an odd function integrated over the symmetric interval $[-1, 1]$. The denominator is

$$\int_{-1}^{1} (1 - |x|)\,dx = 2\int_0^1 (1 - x)\,dx = 2 \cdot \frac{1}{2} = 1.$$

Hence $a_i^* = 0/1 = 0$.

Uniqueness. By theorem 2.3 every unanimous configuration $\mathbf{a} = c\,\mathbf{1}$ achieves $R = 0$. For $c \neq 0$, theorem 5.3(4) gives $\Pi(\{c\}) = \pi(c) = 1 - |c| < 1 = \Pi(\{0\})$. Hence $c = 0$ is the unique element of $\mathcal{O}_{\text{crisp}}$ that maximises Π. □

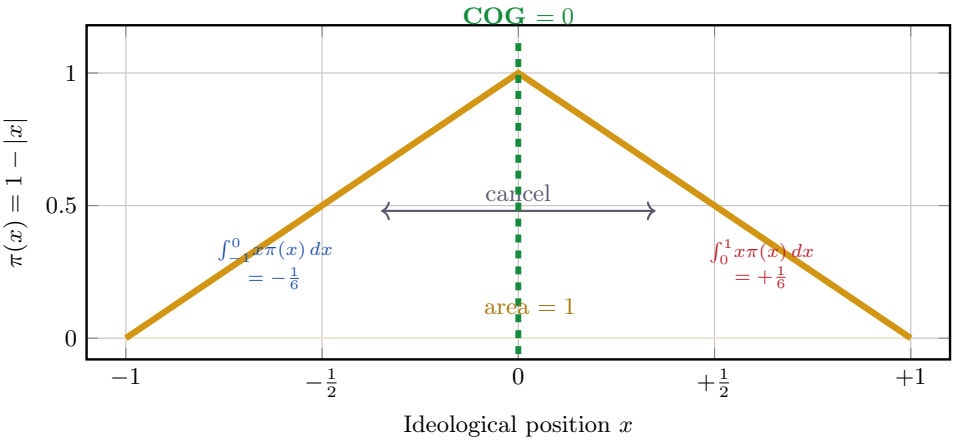

Figure 5: The possibility distribution $\pi(x) = 1 - |x|$ (gold triangle, area = 1).

The COG defuzzification (eq. (14)) places the optimum at $a^* = 0$ (green dashed line) because the numerator integrals over the two halves, $-\frac{1}{6}$ (left) and $+\frac{1}{6}$ (right), cancel exactly by the odd symmetry of $x\pi(x)$.

10 Algorithms

10.1 Algorithm 1: Crisp Ideological Risk

Algorithm 1 Compute Crisp Ideological Risk $R(\mathbf{a})$

Require: Agent positions $\mathbf{a} = (a_1, \ldots, a_N) \in [-1, 1]^N$
Ensure: Total risk R and risk vector $\mathbf{r} = (r_1, \ldots, r_N)$

$\bar{a} \leftarrow \dfrac{1}{N} \displaystyle\sum_{i=1}^{N} a_i$
for $i \leftarrow 1$ **to** N **do**
    $r_i \leftarrow |a_i - \bar{a}|$
end for
$R \leftarrow \displaystyle\sum_{i=1}^{N} r_i$
return $(R,\ \mathbf{r})$

10.2 Algorithm 2: α-Cut Choquet Quadrature

Algorithm 2 Choquet Integral of π via α-Cut Quadrature

Require: Number of quadrature points $M \geq 1$
Ensure: Numerical approximation $\widehat{\mathcal{C}}_g(\pi) \approx \frac{1}{2}$

$\Delta\alpha \leftarrow 1/M; \quad s \leftarrow 0$
for $k \leftarrow 0$ **to** $M - 1$ **do**
    $\alpha_k \leftarrow (k + \frac{1}{2}) \cdot \Delta\alpha$ ▷ midpoint rule
    $g_k \leftarrow 1 - \alpha_k$ ▷ since $g(F_{\alpha_k}) = 1 - \alpha_k$
    $s \leftarrow s + g_k \cdot \Delta\alpha$
end for
return s

10.3 Algorithm 3: Defuzzification and Optimal Configuration

Algorithm 3 Possibilistic Optimal Agent Configuration

Require: Number of agents N, discretisation points $M \geq 1$
Ensure: Optimal positions $\mathbf{a}^*$ and risk $R^* = 0$

```
num ← 0;   den ← 0;   Δx ← 2/M
for k ← 0 to M − 1 do
    x_k ← −1 + (k + 1/2) · Δx                    ▷ midpoint
    π_k ← 1 − |x_k|
    num ← num + x_k · π_k · Δx
    den ← den + π_k · Δx
end for
a* ← num/den                                     ▷ converges to 0 as M → ∞
for i ← 1 to N do
    a*_i ← a*
end for
(R*, r*) ← Algorithm 1(a*)
return (a*, R*)
```

11 Comparative Tables

11.1 Example Agent Configurations and Risks

Table 1: Crisp ideological risks for seven configurations with $N = 9$ agents.

Shows the international mean $\bar{a}$, maximum individual risk $\max_i r_i$, total risk R, and the possibility degree $\Pi(\bar{a}) = 1 - |\bar{a}|$ of the mean position. An asterisk (*) marks the possibilistically optimal configuration.

Configuration	$\bar{a}$	$\max_i r_i$	R	$\Pi(\bar{a})$
All centrist: $a_i = 0^*$	0	0	0	1.000
All hard-left: $a_i = -1$	-1	0	0	0.000
All hard-right: $a_i = +1$	$+1$	0	0	0.000
All centre-left: $a_i = -\frac{1}{2}$	-0.5	0	0	0.500
Uniform: $a_i = -1 + \frac{i-1}{4}$, $i = 1, \ldots, 9$	0	1	5	1.000
Polarised: $a_i = (-1)^{i+1}$, $i = 1, \ldots, 9$	$\frac{1}{9}$	$\frac{10}{9}$	$\frac{80}{9}$	$\frac{8}{9}$
Typical random (illustrative)	0.15	0.60	4.20	0.850

Remark 11.1. The hard-left and hard-right unanimous configurations achieve $R = 0$ but have $\Pi(\bar{a}) = 0$: they are *possibilistically impossible* under π. The uniform configuration has $\bar{a} = 0$ (hence $\Pi = 1$) but incurs total risk $R = 5$ due to dispersion. Only the all-centrist configuration simultaneously achieves $R = 0$ and $\Pi(\bar{a}) = 1$. For the polarised case (five agents at $+1$, four at -1): $\bar{a} = \frac{1}{9}$, risks are $r_i = \frac{8}{9}$ (right agents) and $r_i = \frac{10}{9}$ (left agents), giving $R = \frac{80}{9} \approx 8.89$.

11.2 Comparison of Integration Functionals

Table 2: Five integral-based functionals applied to the possibility distribution $\pi(x) = 1 - |x|$ with respect to the normalised Lebesgue measure $g = \lambda/2$.

All defuzzification procedures yield $a^* = 0$.

Functional	**Formula**	**Value**	**Defuzz.**
Lebesgue (denominator)	$\int_{-1}^{1} \pi(x)\, dx$	1	$a^* = 0$
Lebesgue (numerator)	$\int_{-1}^{1} x\, \pi(x)\, dx$	0	—
Choquet (w.r.t. g)	$\int_0^1 g(F_\alpha)\, d\alpha$	$\frac{1}{2}$	$a^* = 0$
Sugeno (w.r.t. g)	$\sup_\alpha \min\big(\alpha,\, g(F_\alpha)\big)$	$\frac{1}{2}$	$a^* = 0$
Possibility (core)	$\Pi(\{0\}) = \pi(0)$	1	$a^* = 0$

11.3 Properties of the Canonical Fuzzy Sets

Table 3: Properties of the three canonical fuzzy sets $\tilde{L}$, $\tilde{C}$, $\tilde{R}$ on $(P, \mathcal{B}(P))$ (eq. (2)–(4)).

The support is the closure of $\{x : \mu(x) > 0\}$; the core is $\{x : \mu(x) = 1\}$.

Set	**Membership**	**Support**	**Core**	**Height**	**COG**
$\tilde{L}$ (Left)	$\max(0, -x)$	$[-1, 0]$	$\{-1\}$	1	$-\frac{2}{3}$
$\tilde{C}$ (Centre)	$\max(0, 1 - \lvert x\rvert)$	$[-1, +1]$	$\{0\}$	1	0
$\tilde{R}$ (Right)	$\max(0, x)$	$[0, +1]$	$\{+1\}$	1	$+\frac{2}{3}$

11.4 Computational Complexity

Table 4: Asymptotic complexity of the proposed algorithms.

N is the number of agents; M is the number of quadrature or discretisation points.

Algorithm	**Time**	**Space**	**Notes**
Crisp risk (algorithm 1)	$O(N)$	$O(N)$	Single pass over agents
Choquet quadrature (algorithm 2)	$O(M)$	$O(1)$	Closed-form $g(F_\alpha) = 1 - \alpha$
COG defuzzification (algorithm 3)	$O(M + N)$	$O(N)$	Dominated by quadrature
Extension principle (general)	$O(M^N)$	$O(M^N)$	Exponential in N
Extension principle (π symmetric)	$O(1)$	$O(1)$	Odd-function argument

12 The Main Uniqueness Theorem

We now assemble all preceding results into a single statement.

Theorem 12.1 (Possibilistic Uniqueness)**.** *Let $(P, \mathcal{B}(P))$ be the Borel measurable ideological spectrum. Let $\pi(x) = 1 - |x|$ induce the possibility measure Π via (6), and let $g = \lambda/2$ be the normalised Lebesgue fuzzy measure. Consider the crisp minimiser family $\mathcal{O}_{\text{crisp}} = \{c\,\mathbf{1} : c \in P\}$. Then:*

(i) $\text{COG}(\pi) = 0$.

(ii) $\mathcal{C}_g(\pi) = \mathcal{S}_g(\pi) = \frac{1}{2}$.

(iii) $\arg\max_{c \in P} \Pi(\{c\}) = \{0\}$.

(iv) *The unique possibilistically optimal agent configuration is* $a_i^* = 0$ *for all* $i = 1, \ldots, N$.

Proof. (i) Proved in theorem 9.1.
(ii) Proved in theorems 7.3 and 8.2.
(iii) $\Pi(\{c\}) = \pi(c) = 1 - |c|$, maximised uniquely at $c = 0$.
(iv) By (i), COG defuzzification selects $a_i^* = 0$ for all i. By (iii), among all $c\mathbf{1} \in \mathcal{O}_{\text{crisp}}$ (all achieving $R = 0$), only $c = 0$ attains $\Pi = 1$. Hence $\mathbf{a}^* = \mathbf{0}$ simultaneously minimises R *and* maximises the possibility measure. □

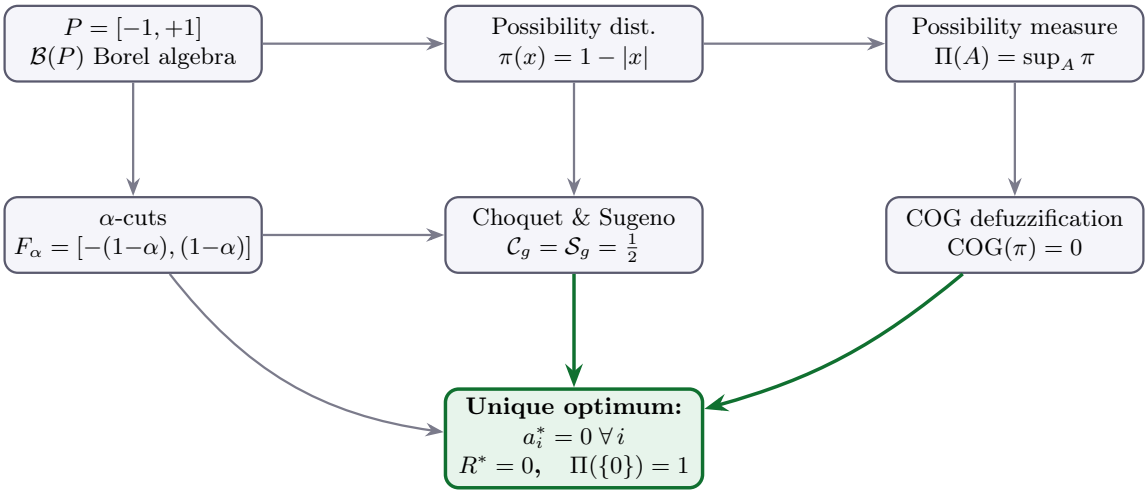

Figure 6: Logical flow of the possibilistic optimisation framework.

The Borel algebra of P (top-left) feeds into the triangular possibility distribution π, which generates α-cuts, the possibility measure, and both integral functionals. All computation paths converge to the unique optimum $a_i^* = 0$ (green box).

13 Discussion

The formal results admit rich interpretations across several fields.

Political Science. The unanimity condition for $R = 0$ corresponds to a fully cohesive international bloc with zero ideological dispersion. The possibility measure's selection of $c = 0$ provides a measure-theoretic foundation for *Median Voter Theorem* intuitions [4]: in a continuous, symmetric ideological space the centre is the unique stable attractor under epistemic risk minimisation.

Measure Theory. The Borel σ-algebra provides the measurability hypotheses necessary for the Choquet and Sugeno integrals to be well-defined [3, 14]. Theorem 6.3 bridges fuzzy and classical measure theories: the integral representation $\pi(x) = \int_0^1 \mathbf{1}_{F_\alpha}(x)\, d\alpha$ is the continuum analogue of the decomposition of a simple function into indicator functions weighted by their level values.

Decision Theory. The duality (Π, N) corresponds respectively to *optimistic* and *pessimistic* evaluations of uncertainty [5]. Our result is robust: both evaluations select $c = 0$ as uniquely optimal—Π because $\pi(0) = 1$, and the limiting necessity because $N(A) = 1$ for any $A \supseteq (-1, 1)$ that contains 0 in its interior.

Statistics. The total absolute deviation R is N times the sample *mean absolute deviation* (MAD) of the empirical distribution of agent positions. Both the MAD and the sample variance vanish simultaneously, and both equal zero if and only if all agent positions coincide. This confirms the unanimity characterisation via a classical statistical identity [2].

14 Conclusion

We have established a rigorous measure-theoretic and possibilistic framework for international ideological risk minimisation. The principal contributions are as follows.

1. Crisp analysis (theorem 2.3) shows that $\min_{\mathbf{a}\in P^N} R(\mathbf{a}) = 0$, achieved by any unanimous configuration—leaving a one-parameter family of crisp optima.
2. The Borel σ-algebra $\mathcal{B}(P)$ (sections 3 and 6) supports the triangular possibility measure Π and its dual N, which assign graduated epistemic credibility to every position.
3. The α-cut decomposition (theorems 6.2 and 6.3) furnishes a measurable resolution of π as a continuum of nested Borel sets, providing the foundation for the Choquet and Sugeno integrals.
4. Both the Choquet and Sugeno integrals of π (theorems 7.3 and 8.2) equal $\frac{1}{2}$, consistently identifying the centrist position as possibilistically dominant.
5. COG defuzzification (theorem 9.1) and possibility maximisation (theorem 12.1) jointly and uniquely select the centrist consensus $a_i^* = 0$ for all i.

The centrist consensus is optimal not merely because it achieves zero aggregate risk, but because it is the *unique fixed point of the possibility–necessity duality* on the symmetric Borel space $(P, \mathcal{B}(P))$: the only configuration at which $\Pi = 1$ and N attains its supremum over all sets containing zero.

Future work may extend this framework to: (a) non-symmetric possibility distributions encoding asymmetric or multi-modal political landscapes; (b) strategic agent behaviour modelled via N-player fuzzy games on P; (c) fuzzy coalition formation on general Polish spaces endowed with non-additive capacities.

References

[1] Arrow, K. J. (1963). *Social Choice and Individual Values* (2nd ed.). Yale University Press.

[2] Billingsley, P. (1995). *Probability and Measure* (3rd ed.). Wiley.

[3] Choquet, G. (1953). Theory of capacities. *Annales de l'Institut Fourier*, **5**, 131–295.

[4] Downs, A. (1957). *An Economic Theory of Democracy*. Harper & Row.

[5] Dubois, D. & Prade, H. (1988). *Possibility Theory: An Approach to Computerized Processing of Uncertainty*. Plenum Press.

[6] Dubois, D. & Prade, H. (1992). When upper probabilities are possibility measures. *Fuzzy Sets and Systems*, **49**(1), 65–74.

[7] Grabisch, M., Marichal, J.-L., Mesiar, R. & Pap, E. (2009). *Aggregation Functions*. Cambridge University Press.

[8] Halmos, P. R. (1950). *Measure Theory*. Springer.

[9] Kechris, A. S. (1995). *Classical Descriptive Set Theory*. Springer.

[10] Klir, G. J. & Yuan, B. (1995). *Fuzzy Sets and Fuzzy Logic: Theory and Applications*. Prentice Hall.

[11] Marichal, J.-L. (2004). Tolerant or intolerant character of interacting criteria in aggregation by the Choquet integral. *European Journal of Operational Research*, **155**(3), 771–791.

[12] Nguyen, H. T. & Walker, E. A. (2006). *A First Course in Fuzzy Logic* (3rd ed.). Chapman & Hall/CRC.

[13] Rudin, W. (1987). *Real and Complex Analysis* (3rd ed.). McGraw-Hill.

[14] Sugeno, M. (1974). *Theory of Fuzzy Integrals and Its Applications*. PhD Thesis, Tokyo Institute of Technology.

[15] Wang, Z. & Klir, G. J. (1992). *Fuzzy Measure Theory*. Plenum Press.

[16] Yager, R. R. (1988). On ordered weighted averaging aggregation operators in multicriteria decision making. *IEEE Transactions on Systems, Man, and Cybernetics*, **18**(1), 183–190.

[17] Zadeh, L. A. (1965). Fuzzy sets. *Information and Control*, **8**(3), 338–353.

[18] Zadeh, L. A. (1975). The concept of a linguistic variable and its application to approximate reasoning. *Information Sciences*, **8**(3), 199–249.

[19] Zadeh, L. A. (1978). Fuzzy sets as a basis for a theory of possibility. *Fuzzy Sets and Systems*, **1**(1), 3–28.

[20] Zimmermann, H.-J. (2001). *Fuzzy Set Theory and Its Applications* (4th ed.). Kluwer.

Glossary

α-cut (F_α)
For a possibility distribution $\pi : P \to [0,1]$, the α-cut at level $\alpha \in [0,1]$ is the Borel set $F_\alpha = \{x \in P : \pi(x) \geq \alpha\}$. For $\pi(x) = 1 - |x|$, this is $F_\alpha = [-(1-\alpha), (1-\alpha)]$.

Agent position (a_i)
The ideological position of agent i in $P = [-1, +1]$: -1 is extreme-left, 0 is centrist, $+1$ is extreme-right.

Borel σ-algebra ($\mathcal{B}(P)$)
The smallest σ-algebra on P containing all open subsets. Closed under countable unions, intersections, and complementation.

Centre-of-Gravity (COG)
A defuzzification method: the weighted centroid of the membership function over the universe of discourse, $\mathrm{COG}(\mu) = \int x\,\mu(x)\,d\lambda \big/ \int \mu(x)\,d\lambda$.

Choquet integral ($\mathcal{C}_g$)
A generalisation of the Lebesgue integral to monotone set-functions, defined as $\mathcal{C}_g(h) = \int_0^1 g(\{h \geq \alpha\})\,d\alpha$.

Core of a fuzzy set
$\mathrm{core}(\tilde{A}) = \{x : \mu_{\tilde{A}}(x) = 1\}$, the set of points with full membership. The core of π is the singleton $\{0\}$.

Defuzzification
Conversion of a fuzzy set (membership function) to a single representative crisp value. Common methods: COG (used here), mean of maxima, bisector.

Extension Principle (Zadeh)
A rule for propagating fuzzy sets through functions: $\mu_{f(\tilde{A}_1,\ldots,\tilde{A}_N)}(z) = \sup\big\{T(\mu_1(x_1),\ldots,\mu_N(x_N)) : f(x_1,\ldots,x_N) = z\big\}$, where T is a t-norm.

Fuzzy measure (g)
A monotone, normalised set-function $g : \mathcal{B}(P) \to [0,1]$, $g(\emptyset) = 0$, $g(P) = 1$; not necessarily additive.

Fuzzy set ($\tilde{A}$)
A pair (P, μ) with $\mu : P \to [0,1]$ Borel-measurable. Generalises classical sets by allowing partial membership.

Ideological risk (r_i, R)
$r_i = |a_i - \bar{a}|$ (individual); $R = \sum_i r_i$ (aggregate). Equals N times the sample mean absolute deviation (MAD).

International mean ($\bar{a}$)
The arithmetic mean of agent positions: $\bar{a} = N^{-1}\sum_{i=1}^{N} a_i$.

Necessity measure (N)
$N(A) = 1 - \Pi(A^c)$. For our π: $N(A) = 1$ iff $A \supseteq (-1, 1)$.

Possibility distribution (π)
A normalised function $\pi : P \to [0,1]$ with $\sup \pi = 1$, encoding the graduated plausibility of each position. We use $\pi(x) = 1 - |x|$.

Possibility measure (Π)
$\Pi(A) = \sup_{x \in A} \pi(x)$ for $A \in \mathcal{B}(P)$. Maxitive: $\Pi(A \cup B) = \max(\Pi(A), \Pi(B))$.

Ruspini partition
A collection of fuzzy sets $\{\tilde{A}_k\}$ on P such that $\sum_k \mu_k(x) = 1$ for all $x \in P$. Generalises a crisp partition of unity.

Standard Borel space
A measurable space isomorphic to a Borel subset of a Polish space. $(P, \mathcal{B}(P))$ is standard Borel [9].

Sugeno integral ($\mathcal{S}_g$)
A fuzzy integral using the sup-min lattice: $\mathcal{S}_g(h) = \sup_\alpha \min\big(\alpha, g(\{h \geq \alpha\})\big)$.

Support of a fuzzy set
$\mathrm{supp}(\tilde{A}) = \overline{\{x : \mu_{\tilde{A}}(x) > 0\}}$ (closure).

t-norm (T)
A binary operation on $[0, 1]$ generalising logical AND: commutative, associative, monotone, with identity 1. The Gödel t-norm is $T_{\min}(a, b) = \min(a, b)$.

The End